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Energy Transition in Ghana: Evidence from Hamburg (Germany) and Los Angeles (USA)

Abstract

Globally, there has been a call to transition from a fossil-fuel-dependent energy system to renewable alternatives. This call is due to the rising impact of climate change, often caused by anthropogenic factors. Countries in the Global North have embarked on aggressive measures to decarbonize, while those in the Global South have not matched this momentum. Developing countries in the Global South often face the foundational challenge of energy access, alongside growing pressure to reduce greenhouse gas emissions. This paper argues that cities in the developing Global South, such as Accra, can pursue this energy transition by following the examples of Hamburg (Germany) and Los Angeles, California (USA). Drawing on a comparative case-study methodology and informed by sociotechnical transition theory and the policy transfer theory, this paper investigates how policy architecture, institutional arrangements, financing mechanisms, citizen-engagement strategies, and technological innovations that have underpinned energy transitions in these leading jurisdictions. The findings show that for a successful transition to occur, there should be a strong legislative framework, decentralized governance, encouragement of public-private partnerships, and, lastly, community and citizen participation. This paper argues that, despite the current socioeconomic and sociopolitical challenges Accra (by extension, Ghana) faces, it can still accelerate its progress towards an energy transition from a fossil-fueled system to renewable sources. This study contributes to the body of knowledge of how experiences in the Global North can help shape urban energy governance.

1. INTRODUCTION

Globally, the world's energy system is undergoing a rapid, phase-scale metamorphosis. The factors driving this metamorphosis include climate change. (Rial, 2024) and also the need for a long-term and sustainable energy supply (Rabbi et al., 2022). Countries in the Global North are primarily moving away from fossil-fuel-based systems to renewable energy sources, such as solar, wind, hydrogen, biofuels, and advanced battery technologies. (Chen et al., 2023). Despite the movement to renewable sources, as of 2024, fossil fuels contribute 73% of global greenhouse gas emissions (GHG) (Yaman, 2024). This prevalence has informed calls for every country, whether in the Global North or South, to move towards renewable and clean energy (Agoundedemba et al., 2023).

Urban cities, by design, are major contributors to GHG emissions, and as a result, they are integral to these energy transitions. Urban cities are epicenters of economic activity due to their relatively high population densities and their function as political agencies. (Noaime et al., 2025). Due to cities being major contributors to GHG emissions, they should be used as vehicles for decarbonization. (Franco et al., 2023). Cities in both the Global North and South have partnered with international bodies such as the United Nations Environment Programme (UNEP) to develop urban climate action plans to implement the goals of the Paris Agreement. The Paris Agreement has been integral in the energy transition efforts of Hamburg and Los Angeles. These cities have developed a unique approach to drive energy transition. These approaches combine policy instruments, institutional reforms, and civic engagement strategies.

These two Global North cities can provide necessary examples for how cities in sub-Saharan Africa can accelerate towards transition to renewable energy sources. Developing countries face significant challenges that impede their move towards renewable energy sources; however, there are still opportunities for their transition. Accra is a perfect example of this duality of challenges and opportunities. The city currently relies heavily on a centralized hydroelectric grid managed by the Electricity Company of Ghana (ECG). ECG is currently plagued with persistent inefficiencies, financial fragility, and limited capacity for low-carbon expansion. Accra finds itself in the tropics, which is an area well-endowed with an untapped solar resource. The area where Accra finds itself receives between 4 and 6 kWh/m^2 of solar irradiation daily and between 1,800 and 3,000 sunshine hours per year (Mensah et al., 2025). Against this background, the Government of Ghana (GOG) launched a Renewable Energy Policy in partnership with Sustainable Energy for All. This signaled a political commitment to diversifying the country's national energy mix as they are unable to translate. Like other African cities, Accra has challenges in energy transitions. However, in Accra, as in other African cities, the challenge lies in translating this national mandate into actionable city-level interventions.

The purpose of this paper is therefore to address the challenges faced by Global South cities, such as Accra, in its energy transition by following evidence from Hamburg and Los Angeles to generate policy-relevant insights for Accra. At its core, this paper asks a fundamental question:

What policy lessons can Accra, Ghana, draw from the energy transition experiences of Hamburg and Los Angeles to advance its own renewable energy agenda?

To answer this question, the paper has four main sections: introduction, conceptual framework, comparative case studies of Hamburg and Los Angeles, and, lastly, a synthesis of lessons for Accra. The last chapter concludes with recommendations and directions for future research.

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2. CONCEPTUAL FRAMEWORK AND METHODOLOGY

Conceptually, this paper is based on two theories: the sociotechnical transition theory and the policy transfer theory. The sociotechnical transition theory is rooted in the multi-level perspective (MLP) developed by (Geels, 2002). This theory posits that energy systems are sociotechnical regimes, they therefore require stable configurations of technologies, institutions, markets, user practices and regulatory norms. These systems much co-evolve over time to enable energy transitions to occur (Radtke, 2025). Primarily, this theory means that for energy transitions to occur, it should be upheld by individuals, the government, and public-private agreements as well as other configurations. The theory aggregates these regimes in three levels: the macro-level landscape (global climate change imperatives and international energy agreements), the meso-level regime (incumbent energy utilities, national grid infrastructure, and available regulatory frameworks), and lastly the micro-level regime (community solar initiative, pilot renewable energy projects, and grassroots civil campaigns). For the transition to occur, the micro-level regime must accumulate grave momentum to destabilize and reconfigure existing regimes, accelerated by landscape pressure. (Bellanca, 2025). This paper, as applied here, is used to interpret how Hamburg and Los Angeles have navigated the disruption of fossil fuel-dependent energy regimes, and to assess the extent to which Accra's nascent renewable energy initiatives constitute viable niche innovations with the potential to challenge its centralised hydroelectric regime.

The second theory, Policy Transfer Theory, was developed by Dolowitz & Marsh, (2000) and later refined by Rose (2005) through the concept of lesson-drawing. The theory helps examine the processes by which knowledge about policies, administrative arrangements, institutions, and ideas in one political setting is used to develop policies in another political setting. (Dolowitz & Marsh, 2000). Rose (2005) distinguishes between the idea of wholesale copying of policies and the more sensitive practice of lesson-drawing in which policymakers extract principles and adapt them to their own institutional, cultural, and political context as applied in Accra, where there is a significant difference in governance capacity, fiscal resources, and regulatory authority between the country's capital, metropolitan assemblies, and the sub-national governments of Germany and the United States. By adopting a lesson-drawing rather than a direct transfer approach, this theory will help to identify the adaptation of enabling principles over the replication of specific policy instruments.

These two theories offer a robust analytical framework. The sociotechnical theory helps illustrate the structural conditions and transition dynamics observed in Hamburg and Los Angeles. The policy transfer theory helped guide the extraction and contextualization of lessons that Accra needs for its energy transition. The data for this study were taken from peer-reviewed academic literature, policy documents, government reports, and authoritative secondary sources. The analysis is further informed by Scopus-indexed empirical literature, synthesized using the dual framework to generate transferable, contextually grounded insights for Accra's energy transition planning.

3. METHODOLOGY

3.1. Research Design

This paper adopted a qualitative comparative case study design approach. This method is appropriate for the investigation of complex, context-dependent phenomenon and one of such is energy transitions (Yin et al., 2018). Comparative case study works by offering an in-depth examination of how related policies, policy architecture, institutional arrangements, and implementation mechanisms work within a specific urban context, while also facilitating systematic cross-case analysis to unearth insights that are transferable to other settings and contexts (Krehl & Weck, 2020).

3.2. Rationale for selecting case study

A most-different system design was used to select the cities of Hamburg and Los Angeles. A most-different systems design is a comparative research method that examines cases with diverse contexts to identify commonalities and causal factors, or systems that are similar across the diverse cases (Wagenaar et al., 2022). The purpose of this method was to select cases that differ across key contexts, continents, and context dimensions yet share the outcome of interest. In this case, both Hamburg and Los Angeles use institutionally distinct strategies to achieve the same energy transition outcome.

Hamburg operates within Germany's federal Energiewende framework which is characterized by a national feed-in tariffs, strong legislative mandates and a governance culture that promotes remunicipalisation and participatory democracy by the people who live in the city. While Los Angeles on the other hand operates within the California's state market-oriented but highly regulated energy space. In Los Angeles's landscape, the state-level Renewable Portfolio Standards, carbon pricing mechanisms and LADWP shape their energy transition trajectory. These two approaches are fundamentally different from each other yet the end goal is energy transition.

The reason for their selection is that both are known market leaders in energy transition. They have a mature policy architecture that can be learned by Global South cities, including Accra, Ghana. Further, since their approaches differ, they offer distinct approaches and transferable skills options that can be compared and chosen. It allows to identify which of the approaches are better in different contexts. Lastly, studying two cities on two continents with two distinct cultures allows insights to draw on different parts of the policy architecture to enable energy transitions.

3.3. Data Source

This study used data from four sources; the first was peer-reviewed academic literature from Scopus-indexed journal articles and scholarly books. The selected literature focused on energy transitions, urban climate change, and other topics related to Hamburg and Los Angeles. The second data source was from official government documents, which relate to climate plans, energy strategies, legislative text and other reports published by the city's authorities. The third data source was from authoritative sources from recognizable organizations such as the United Nations Environment Program, the International Energy Agencies, among others. The final data source was from empirical transition literature, especially those that talk about and apply sociotechnical transition theory and policy transfer framework. This data source was used to help discuss the theoretical framework of this study.

3.4. Analytical Framework

The data gotten from the data sources were analyzed through the socio-technical transition theory and the policy transfer framework. These two theories are what underpin this study, and hence, the data gathered were filtered through them. Geel's MLP helped interpret how Hamburg and Los Angeles have navigated the disruption of fossil-fuel-dependent energy regimes. In each case, this study identified the relevant landscape pressures, regime characteristics, and niche innovations. The second theory, Policy Transfer Theory, helped in identifying workable systems, policies and principles that can be transferred to Accra. This theory was helpful in the identification of outcomes that is likely to work in the context of Accra.

3.5. Limitation

This study admits certain limitations. Firstly, this study relies extensively on secondary data and this may introduce inherent biases and gaps that already exists in that literature. Further, if primary data were taken, it would have enabled the researcher to identify more nuanced solutions that could have better aided Accra's energy transition. Furthermore, the study did not take into consideration the socio-political-economic landscape of Accra that mitigates against energy transitions in Accra. Lastly, the study admits that the Global North and South are fundamentally different and therefore drawing parallels to them may not be adequate. However, this was done to serve as a source of inspiration to Global South countries about what they can achieve should they learn from the Global North.

4. CASE STUDIES IN ENERGY TRANSITION

4.1 Hamburg, Germany

4.1.1 Context and Climate Vulnerability

The first city of interest is Hamburg. This city is located on the Elbe River in northern Germany. The city is home to 1.8 million inhabitants and has one of the largest ports in Europe (Strupp, 2024). Some of the challenges that this city faces are rising sea levels, more frequent and intense precipitation events, and increasing temperatures associated with anthropogenic climate change (Hanf et al., 2024). These risks have motivated the city to adopt both mitigation and adaption strategies that focuses on energy transitions as a way of solving these challenges. The ports in Hamburg are known for generating considerable CO_2 emissions from their industrial and commercial activities. This grave emission has forced the city to develop a site of green innovations, with some container terminals already utilizing industrial waste heat and piloting solar energy harvesting systems (Cavalcanti et al., 2024)

4.1.2 Policy Trajectory

Hamburg's energy transition is embedded in Germany's national Energiewende (energy transition) framework. This framework has been shaped by decades of evolution in federal policy. Germany's shift away from fossil fuels and nuclear energy began in the 1970s oil crisis and continued in the wake of the 1986 Chornobyl disaster. (Renn & Marshall, 2016). These two incidents galvanized public negativity towards nuclear power across Europe. (Insch & Loughran, 2022). Another landmark milestone was the Renewable Energy Sources Act (Erneuerbare-Energien-Gesetz, EEG) of 2000, which introduced a feed-in tariff (FIT) system that incentivised households and businesses to install solar photovoltaic (PV) systems. The policy offered premium prices for electricity fed back into the national grid. (Fitzgerald, 2020). This policy helped to move Germany's renewable energy share from approximately 3% in 1990 to 35% as of 2020, with a national target of 100% renewable energy by 2045.

At the city level, Hamburg crafted its inaugural Climate Plan in 2015, with a revision done in 2023. The revised plan is now known as the Hamburg Climate Action Law. It established a target of net-zero CO_2 emissions by 2045, five years ahead of the national deadline. The law directed districts to derive 65% of their energy from renewable sources by 2024. It also instructed all new residential buildings to integrate solar PV systems from 2027. Furthermore, it dictated that all government buildings are to achieve carbon neutrality by 2030, and coal-fired heating systems are to be phased out entirely by the same year (Köhnke et al., 2023). This policy directed the transportation sector to promote electric vehicles, expand cycling infrastructure, and promote improved public transit. (Byrne et al., 2021).

Hamburg's policy trajectory has been through a citizen-lead activism. This was achieved through the city's remunicipalization of its privatised electricity, gas, and district heating networks. For instance, following the privatization of the city's utilities in the early 2000s, a grassroots campaign dubbed "Our Hamburg - Our Grid' initiative" successfully mobilized public support for a 2011 referendum mandating the city to repurchase its energy infrastructure (Meyer, 2021). Through this referendum, the city government gained the institutional leverage needed to align energy infrastructure with climate objectives. This led to the establishment of the Hamburg Energie, the municipal renewable energy company.

4.1.3 Implementation Mechanisms

To implement this strategy, Hamburg has used a five-interconnected mechanism. Firstly, the city has leveraged federal policy frameworks, including the EEG feed-in tariff, nuclear phase-out legislation, and the Energiewende target to enable conditions for local action. (Kemmerzell, 2022). The second has been achieved through citizen engagement, institutionalizing remunicipalization, and participatory governance mechanisms that foster a sense of community ownership of the energy transition. The third mechanism is through committed municipal leadership that has driven proactive investment decisions, such as applying for federal funding to develop geothermal energy plants and exploring industrial waste heat for district heating (Yanarella, 2020). The fourth way was when the city cultivated a vibrant clean energy cluster, comprising nearly 200 firms across the wind, solar, and biogas sectors, in collaboration with companies such as Siemens Gamesa Renewable Energy (Elisa, 2024). Finally, the Hamburg University of Technology has been a valuable partner in energy research and innovation. Taken together, these five mechanisms reflect a systemic, multi-actor approach to energy governance that extends well beyond the remit of any single institution.

4.2 Los Angeles, USA

4.2.1 Context and Institutional Architecture

The second setting is Los Angeles, the second most populous city in the United States. It is also one of the significant economic, entertainment and culture hub globally. Los Angeles can be found in California, which is one of the world's largest economies and a state with a long-standing tradition of environmental leadership. Los Angeles' energy system is administered by the Los Angeles Department of Water and Power (LADWP), the largest municipal utility in the United States. The services of LADWP are available to nearly 4 million inhabitants of Los Angeles and it is an important actor in the city's decarbonization efforts (Sheinberg et al., 2025). California's energy and climate policies are managed at the state-level by regulatory bodies such as the California Air Resources Board (CARB), the California Public Utilities Commission (CPUC), and the California Energy Commission. These institutions together comprise one of the most institutionally complex sub-national governance structures for overseeing energy regulations in the state and world at large (Kaswan & Gonzalez, 2025). This state-level framework provides an enabling policy environment within which Los Angeles can pursue its municipal energy transition. This is a structural parallel to how Hamburg operates within the Germany's Energiewende.

4.2.2 Policy Trajectory

Historically, California's energy transition dates to the mid-1990s, when the state restructured its electricity market through Assembly Bill 1890. This led to the introduction of competitive elements into power generation and, therefore, laid the foundation for renewable energy incentives (Schuparra, 2024). In the early 2000s, the California Solar Initiative was launched, offering financial rebates for residential and commercial solar PV installations, leading to significant growth in distributed solar generation. (Sun & Sankar, 2022a). The state also tightened its Renewable Portfolio Standards, with targets rising from 20% by 2017 to 33% by 2020 (Deshmukh et al., 2023).

The passage of the Senate Bill 100 (SB 100) in 2018 created a signal of California's commitment to attaining 100% carbon-free electricity by 2025 (Hotchkiss et al., 2022). The State further enacted the Low Carbon Fuel Standard and the Advanced Clean Trucks regulations to extend decarbonisation beyond the electricity sector into transportation and industry. (Giuliano & Dexter, 2023). Lastly, the State recognized the need to implement interventions to reduce the prevalence and severity of climate-change-driven wildfires. (Franklin & MacDonald, 2024). These wildfires tend to affect grid reliability and can impact investment in energy storage.

4.2.3 Implementation Mechanisms

The city employed a distinct set of implementation mechanisms. The most prominent of them is the use of Renewable Portfolio Standards (RPS) to achieve its decarbonization agenda. RPS is used as a way to drive sustained investment in utility-scale renewables through mandatory procurement requirements (Madadizadeh et al., 2024). There was also the formation of the California Solar Initiative and related programmes that deployed financial subsidies to lower the upfront cost of solar PV adoption for households and businesses. (Sun & Sankar, 2022b). The state further engaged in aggressive building codes that reduced the demand-side energy consumption, while demand-side management programmes have incentivised behavioural change among consumers.

The state also pursued public-private partnerships to scale renewable energy capacity. They collaborated with utility companies and independent power producers on major solar and wind projects. The state invested heavily in energy storage technologies, such as lithium-ion batteries, to manage the intermittency of renewable generation and ensure grid stability (McNamara et al., 2022). There is also the Community Choice Aggregation (CCA) programme, which has led to governance innovation, enabling local governments to procure renewable energy on behalf of their communities (Dokk Smith et al., 2023). This was done to circumvent the inertia of incumbent utilities. Finally, inter-agency coordination among CARB, CPUC, and CEC has ensured policy coherence and minimised regulatory fragmentation across the energy and environment domains.

5. LESSONS FOR ACCRA, GHANA

5.1 The Accra Energy Context

Happenings and lessons from the two cities of Hamburg and Los Angeles can inform Ghana's energy transition policies. The country's statistical agency, the Ghana Statistical Service, indicates that the city has nearly 10% of the country's population (approximately 3 million inhabitants). As the biggest city in Ghana, and one of the biggest in West Africa, it serves as Ghana's and also West Africa's political, economic, and cultural capital. The city's energy system has been described as chaotic, with significant structural vulnerabilities (Yazdanie et al., 2024). The city relies heavily on the Akosombo Hydroelectric Dam, which was constructed in 1964 on the Volta River. (Agodzo et al., 2023). Accra's reliance on hydroelectric dams means it is susceptible to rainfall and other climate-induced hydrological variability. (Ndehedehe, 2022). The country's energy distribution is managed by the state-owned entity, Electricity Company of Ghana (ECG). ECG is currently facing ever increasing financial losses due to infrastructure deficits and high aggregate technical and commercial losses (Osei-Gyebi & Dramani, 2024). There have been attempts to reduce ECG's financial losses through privatization; however, these have largely been unsuccessful.

In September 2023, the country's central government launched a National Renewable Energy policy in partnership with Sustainable Energy for All (Bokoro & Kyamakya, 2024). This was a means to diversify the country's energy mix. Pilot projects were implemented at the Bui Hydropower Station and at the Volta River Authority's satellite facilities in the Upper East Region of Ghana. Despite the country receiving high levels of sunlight, Accra's solar generation remains low and is now attracting government attention. As already indicated, Accra receives between 4 and 6 kWh/m² daily and 1,800 to 3,000 annual sunshine hours (Mensah et al., 2025). Despite its abundant sunshine, the solar industry remains largely untapped. It has the potential to meet the country's energy needs and reduce the country's overdependence on the ECG grid by more than 50% (Odoi-Yorke et al., 2025). The development of Accra's solar industry will also contribute to global efforts to reduce carbon footprints (Amo-Aidoo et al., 2022).

5.2 Policy Lessons from the Comparative Analysis

5.2.1 Decentralization and Citizen Engagement

One important lesson that Accra can learn from Hamburg's Energiewende is that it can achieve energy transition through citizen engagement, involvement, and democratic participation in energy governance. (Fraune, 2022). One thing that Accra can do is to foster a sense of community participation in energy decision-making. This can help in encouraging societal acceptance of renewable energy projects. This can help generate grassroots demand for clean energy solutions and hold both public and private actors accountable for energy transition commitments.

The first theory of sociotechnical transition provides evidence that Hamburg Energie, a niche innovation, has successfully challenged an incumbent energy regime. In this same theoretical perspective, we can posit that community-level initiatives such as solar cooperatives, community energy audits, or citizen-led advocacy for a decentralized grid infrastructure can lead to broader systemic change within the country's centralized hydroelectric energy regime. Drawing on Rose's (2005) lesson-drawing framework, Accra can formalize community participation, as in Hamburg, in metropolitan energy planning through existing institutional channels, such as the various municipal and district assemblies, unit committees, and General Assemblies, which already have some constitutional power for participatory local government. Through the Hamburg example, we can posit that it would serve as a source of inspiration for developing a contextually appropriate governance innovation. Accra should not just copy the Hamburg example; it should use it as an inspiration for its energy transition.

5.2.2 Holistic and Cross-Sectoral Planning

Lessons from the Los Angeles approach to the energy transition indicate that it focuses on integration across multiple sectors, including electricity, transportation, buildings, and industry. (Goedeking & Meckling, 2024). Following Los Angeles's example, Accra's energy transition planning should adopt a cross-sector perspective. Accra should recognize that decarbonization in the energy sector alone is insufficient unless it also involves other sectors. For instance, if transportation and buildings continue to rely on fossil fuels, an energy transition would not be achieved. The various municipal and district assemblies, such as the Accra Metropolitan Assembly, Ga North Municipal Assembly, and Ga South Municipal Assembly, among others, can pursue inter-departmental coordination mechanisms that align planning with the transportation and environmental portfolios behind a commonly agreed-upon climate action agenda. There can be the creation of a dedicated Climate office at various municipal and district assemblies to oversee and report on progress across all relevant actors.

5.2.3 Public–Private Partnerships

Lessons from both Hamburg and Los Angeles demonstrate the need for public-private partnerships in accelerating energy transitions. For instance, Hamburg's renewable energy cluster of almost 200 firms and Los Angeles's utility-developer partnerships for large-scale solar and wind projects are examples of how public-private partnerships can be used as tools for energy transitions. Accra can also promote structured public-private partnerships in solar energy development, especially for residential rooftop solar and mini-grid systems in urban and peri-urban areas. This could help attract both domestic and international investment and leverage private-sector efficiency. Private-public partnerships should be as transparent as possible to ensure competitive tendering processes and a clean regulatory environment. These are all essential in building investor confidence in Accra's market.

5.2.4 Innovative Financing Mechanisms

Another lesson from these two cities is on the role of innovative financing mechanisms in driving energy transitions. A key constraint on the energy transition, especially in developing cities such as Accra, is the availability of affordable finance (Oppong et al., 2023). Los Angeles and Hamburg are wealthy cities that, on their own, can attract significant investment. Accra can explore a range of innovative financing instruments, such as feed-in tariffs, to encourage distributed solar generation. The various municipal and district authorities, as well as the GOG, can look into issuing green bonds to fund renewable energy infrastructure. The city, through the GOG, can explore concessional loans from developmental banks such as the African Development Bank and the World Bank to drive this initiative. The Government of Ghana's existing engagement with Sustainable Energy for All provides a platform for mobilizing additional international climate finance, including from the Green Climate Fund and bilateral donors committed to the energy transition.

5.2.5 Proactive Regulatory Framework

Lessons from both cities highlight the need for a proactive regulatory framework. A central lesson from both cities is that an ambitious and clearly articulated regulatory framework is an essential driver of private-sector behaviour and investment. Accra can take a cue from this by setting up a Metropolitan Energy Transition Plan that outlines clear, binding targets for renewable energy deployment, energy efficiency improvements, and reductions in carbon emissions. This would set out a clear policy direction for investors, businesses, and the residents. The various municipalities and district assemblies can enact bylaws that force or encourage the integration of solar PV systems on new commercial and large residential developments. This strategy is consistent with how Hamburg used to attain energy transitions by targeting new constructions. Accra should streamline permit processes for small-scale solar installations to further reduce barriers to entry for households and businesses.

5.2.6 Leadership, Adaptability, and Political Commitment

A major lesson from these two cities is the role of leadership, adaptability, and political commitment in promoting the energy transition. Political leadership, as exemplified by Hamburg and Los Angeles, is integral to driving large-scale system transitions. Without effective leadership and political commitment, an energy transition cannot happen. Accra can lead from both of these cities. Citizens can play an active role in this: they can enforce political commitments by voting out politicians or governments that are not committed to ensuring the widespread adoption of clean energy. They should also challenge and change the way municipal assembly heads are elected. They should change the law that mandates that the central government elect leaders to these assemblies. The best-case scenario is that these leaders must be elected and, once elected, can push them to commit to a clean energy transition.

6. DISCUSSION

The comparative analysis in this paper tells the story of the facilitators who promote energy transitions in the German city of Hamburg and the USA city of Los Angeles. In the context of these two cities, the facilitators are ambitious and legally binding climate targets set by relevant institutions, the creation of institutions that support policy coordination across multiple agencies and sectors, the creation of mechanisms through which citizens and the community can participate in this energy transition, the creation of a support environment that encourage private-sector innovation and investments and lastly, an active and responsive political leadership. These facilitators can be adapted for the Accra context to drive energy transitions.

The paper makes contributions to the literature on urban energy governance and to lessons that developing countries can adapt to drive energy transitions. By examining the context from the Global North, the paper connects findings from established transition leaders to what developing countries in the Global South can use to avoid failing in their energy transition efforts. Also, by using the theories of sociotechnical transition and policy transfer, this paper developed an analytical tool for cross-contextual comparison and for how countries can learn from established players by not just copying them but adapting policies to fit their local context. The paper recommends policy-relevant agenda for Accra by recommending that they move beyond non-specific solutions to more specific institutional, financial and regulatory frameworks that can help this energy transitions to occur effectively.

The limitation of this paper is that it relies heavily on secondary data from published papers, and future studies should consider primary data from Accra to assess the political feasibility and the current capacities of institutions to implement these proposed solutions. The primary data should primarily examine barriers to the energy transition. Ghana's political economy is complex, as it includes the entrenched interests of state utilities, fuel importers, and political elites, and this paper did not have the capacity to examine these complexities. Lastly, it is essential to look at how Global South countries such as Rwanda, Kenya, and Cape Town have achieved energy transitions, and use them to advise Accra on how to do so, since these cities are more similar to Accra than Hamburg and Los Angeles.

7. CONCLUSION

The paper examines energy transition experiences in Global North regions such as Hamburg and Los Angeles, and how Ghana can draw on them in its own energy transition. The paper establishes that both Global North cities achieved progress in decarbonizing their energy system through a combination of binding legislative frameworks, decentralized governance, citizen engagement, public–private partnerships, innovative financing, and committed political leadership. Despite the hindrances Accra faces, such as limited fiscal capacity, centralized energy governance, inadequate grid infrastructure, and restricted sub-national legislative authority, it also has renewable energy assets, including a growing civil society and a national policy framework that supports energy diversification.

This paper's central argument is that Accra does not need to wait for a national-level solution to begin its own energy transition. As a capital city, it can lead in innovation so that other cities such as Kumasi, Takoradi, Tamale, Lagos, Abuja, and Abidjan can follow. Accra can leverage the institutional mandates of the various municipal and district assemblies to engage communities as active participants in energy governance. The city should also foster public-private investment in distributed solar generation. The city should develop a Metropolitan Energy Transition Plan with clear, measurable targets that can be verified by all and sundry. It must be noted that, due to differences in the systems between Accra, Hamburg, and Los Angeles, Accra should not blindly copy the two cities, but rather use them as sources of inspiration for locally grown innovation.

Although Accra contributes less to GHG emissions, it has a duty of care to help reduce GHG emissions through energy transitions. Fighting GHG is an all-inclusive efforts and Accra's success can motivate other SSA cities to pursue ambitious urban climate action. This paper recommends further research that looks at how barriers such as the political economy constraints and community perceptions mitigate against Accra's energy transition.

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